

Cybersecurity d'Amore (6 of 9 ECTS(CFU)) - 2024-25 syllabus in progress									
NEW: CRITICAL TOPICS ARE HIGHLIGHTED									
Failure to pass critical topics may result in immediate termination of the exam, at the discretion of the teacher									
lect. #	date	classroom	start	end	hours	subject	material	comment	total hours d'Amore
1	25-9-2023	A1 - via Caserta 6	14:00	16:00	02:00	Introduction. Details on the course (web site, homeworks, classroom, examination, textbook, tools). Information security requirements (CIA, authentication, non-repudiation). Introduction to open security, security by obscurity, security by patching, model of encryption. Symmetric encryption. Adversary. Brute-force.	openssl: https://openssl-library.org/source/index.html iptables: https://www.netfilter.org/pub/iptables/	Happy semester everyone	2:00
2	26-9-2023	29 - San Pietro in Vincoli	14:00	16:00	02:00	Still brute-force. Spoofing, flooding. Threat and exploit. Passive and active adversary. Kerckoff's principle. Caesar's cipher. Alphabetic substitution. Frequency analysis attack. Perfect ciphers. Properties. OTP and its properties. Requirement to be perfect. On keystream reuse. Shannon's theorem.			4:00
3	30-9-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	Proof of Shannon's theorem, Attack models, computational power, big numbers. Stream ciphers synchronous and asynchronous. A5, RC4. RC4: generation of permutation and encryption. Bloch ciphers. History. DES, 3-DES RC5 and NIST competition. Rijndael. Congruence. Quotient set and multiplicative group.			6:00
4	2-10-2024	online	14:00	16:00	02:00	Again quotient set and multiplicative group, Totient number and Euler's theorem. Bezout identity and multiplicative inverse. Examples. Galois fields. Polynomials over finite fields. Irreducible polynomials. Implementation of the GF(p^k) arithmetic. Example (even with Maple). AES specifications and rounds.			8:00
	3-10-2024	online	14:00	16:00	02:00	Summary on Rijndael/AES. Animation. Block modes of operation. ECB, CBC, ciphertext stealing, CFB. Ways to analyze a mode of operation: parallelizability, preprocessing, propagation of errors.	Animation of AES: https://youtu.be/gP4PqVGudtg?si=Xmqv479YrTn7vyu2		10:00
	7-10-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	OFB, CTR. Key whitening. 3-DES EEE, EDE. Meet in the middle.			12:00
	9-10-2024	G - via Caserta 6	14:00	16:00	02:00	Integrity. Model, adversary. CBC MAC. Hashing. Collision resistance. Cryptographic hashing. Birthday paradox and attack. Unkeyed/keyed hashing.			14:00
	14-10-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	Keyed hashing. How to transform an unkeyed hash function to a keyed hash function. Merkle-Damgard construction. Discussion on key as a prefix or as a suffix. Some hashing functions. Overview of SHA-1. Dismission of SHA-1. HMAC. The birthday bound for HMAC. AE and AEAD. EtM, MtE, E&M.			16:00
	16-10-2024	G - via Caserta 6	14:00	16:00	02:00	GCM. Introduction to Diffie-Hellman model: public key cryptography. HW02.			18:00
	17-10-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	Discrete log. Fast exponentiation. Modular properties. One-way function and DL as one-way function. DH public exchange of keys. Perfect forward secrecy. Property of DH and vulnerability to MITM. The product of two big primes p and q. The size of Z_{pq}^* .		one third!	20:00
	23-10-2024	G - via Caserta 6	14:00	16:00	02:00	Exponentiation in Z_{pq}^* . RSA cryptosystem. Proof on decryption. RSA example. Computation of the multiplicative inverse. Bezout identity and Euclidean extended algorithm. Construction of an RSA instance and implementation.			22:00
	24-10-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	RSA as a collection of trapdoor one way functions. Attacks on textbook RSA: factoring, special cases, small e, through CRT (Chinese remainder theorem), small message space, sam,e N, multiplicative property and chosen ciphertext attack, energy based. Summary. PKCS#1 v1. PKCS in general and PKI. OAEP: schema.			24:00
	28-10-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	All or nothing security of OASP. Modern PKCS#1.RSAES-OAEP. OS2IP and I2OSP with examples. Again RSA with old standard: RSAES-PKCS1-v1_5: padding format, example in bash with pkeyutl. Overview. Comparison 1.5 vs. 2.2 (OAEP). ElGamal encryption.		Also in streaming	26:00
	30-10-2024	G - via Caserta 6	14:00	16:00	02:00	Why MAC does not support non-repudiation. Types of forgery. Digital signature definition and verification.RSA signature. PKCS#1 v1.5. Step by step (even in openssl).			28:00

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	31-10-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	EIGamal signature. Key generation. Signining. Verification. Security. Mathematical correctness. RSA vs. ElGamal. DSS and DSA. Keys and signature. Verification and correctness. DSS security. If attacker knows some secret information. Efficiency. DSS vs. RSA. DSA vs. RSA.		one half!!	30:00
	04-11-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	Random numbers: RNG, PRNG and CSPRNG. Cryptographic quality. Difference between cryptographic quality and randomness. Generation of randomness. Initial seed and common mistakes. Randomness in Netscape 1.1 and in Debian. BSI criteria. Definition of cryptographic quality. Four algorithms. Considerations and discussion.		rand and srand in C	32:00
	06-11-2024	G - via Caserta 6	14:00	16:00	02:00	Authentication, definition closed world assumption, negation as failure. Authentication in a closed environment based on a trusted third party. Human vs computer authentication, authentication of people. Replay attack, transmission of credentials encrypted or hashed. Authentication based on public keys. Protocol from Needham-Schröder			34:00
	07-11-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	Attack (reflection attack) to NH. Fix. X509; nonces, timestamps and their use. Digital certificates. One-, two-, three-ways authentication and discussion on the approaches. Challenge-response. Canadian attack. PKI. Fields of a certificate. X500 names. Certification authorities. Examples.	https://www.agid.gov.it/it/piattaforme/firma-elettronica-qualificata/software-verifica		36:00
	11-10-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	Digital certificated in practice. CRLs:meaning and limits. Significance. Fields. Discussion. OCSP. OCSP stapling. PAdES, CAdES, XAdES, JAdES. Details on PAdES: practical examples.			38:00
	13-11-2024	G - via Caserta 6	14:00	16:00	02:00	CAdES and P7M, more practical examples, local/remote signature, verification completed (policies)		two thirds	40:00
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	14-11-2024	29 - San Pietro in Vincoli	14:00	16:00	02:00	more examples of signature. Recap on verification. Still on authentication			42:00
	18-11-2024	29 - San Pietro in Vincoli	12:00	14:00	02:00	Kerberos		switch with Querzoni	44:00
		G - via Caserta 6	14:00	16:00	02:00	Passwords			46:00
		29 - San Pietro in Vincoli	14:00	16:00	02:00	IPsec			48:00
		29 - San Pietro in Vincoli	14:00	16:00	02:00	TLS			50:00
		G - via Caserta 6	14:00	16:00	02:00	SSH			52:00
		29 - San Pietro in Vincoli	14:00	16:00	02:00	Firewalls 1			54:00
		29 - San Pietro in Vincoli	14:00	16:00	02:00	Firewalls 2			56:00
		G - via Caserta 6	14:00	16:00	02:00	Shamir SS			58:00
		29 - San Pietro in Vincoli	14:00	16:00	02:00	Exam texts and recap			60:00